A Simulation-Based Comparative Analysis of Serverless and Provisioned Cloud Relational Databases Under Burst Analytics Workloads

Author TBD[1*]

[1*]Department of Information Technology, TBD (pending owner confirmation), TBD, TBD.

Corresponding author(s). E-mail(s): tbd@example.org;

Abstract

Keywords: serverless databases, provisioned databases, burst workloads, discrete-event simulation, cost model, PostgreSQL

1 Introduction

2 Related Work

3 System and Workload

4 Simulator and Calibration

5 Policies and Cost Model

6 Validation

7 Results

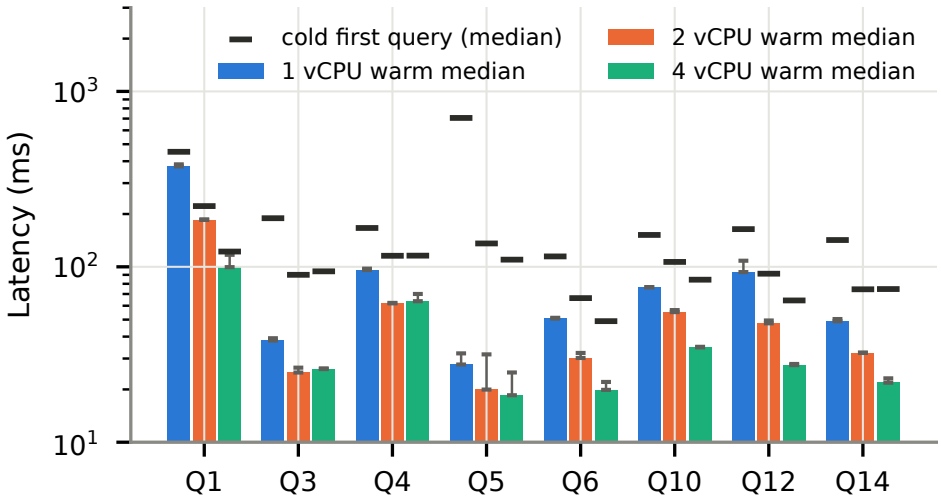


Fig. 1 Placeholder caption. Source: `results/calib_docker.json`.

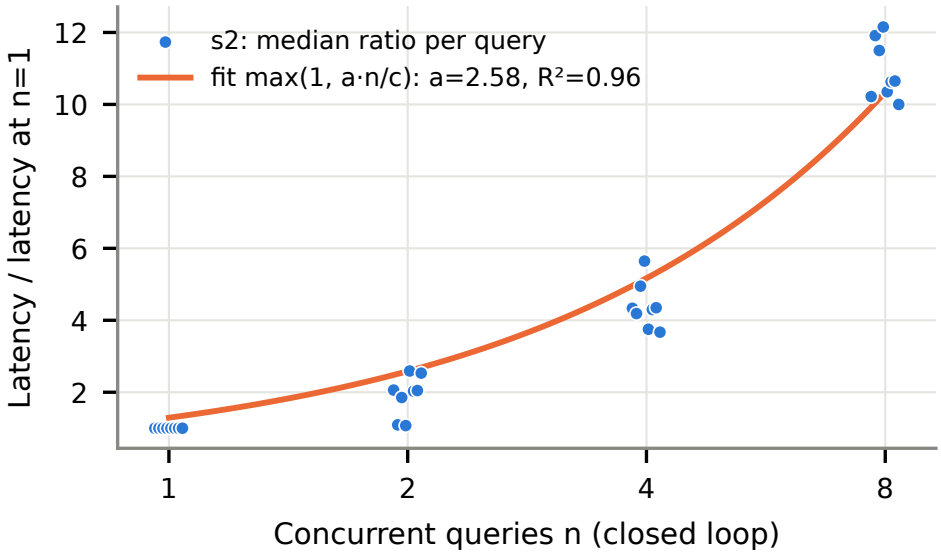


Fig. 2 Placeholder caption. Source: `results/calib_contention.json`.

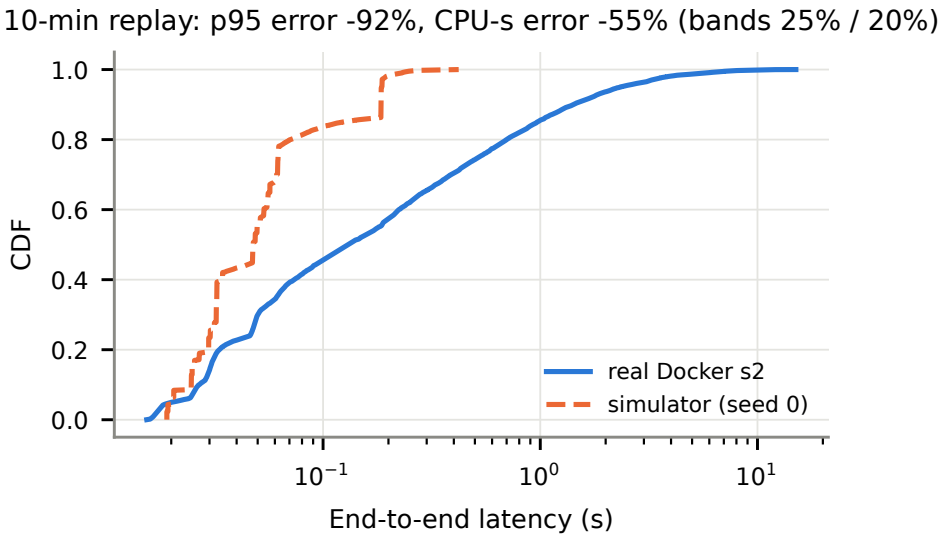


Fig. 3 Placeholder caption. Source: `results/validation.json`.

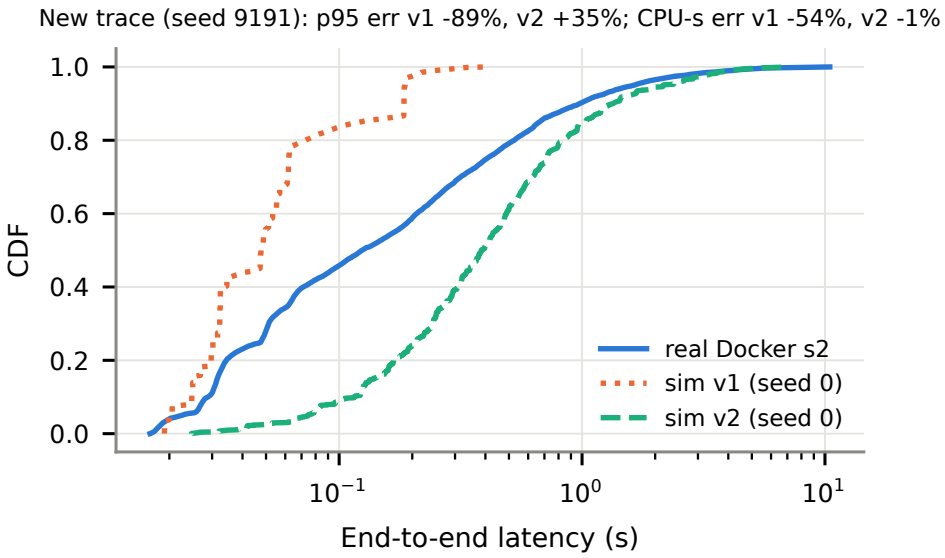


Fig. 4 Placeholder caption. Source: `results/validation_v2.json`.

Table 1 Docker PostgreSQL calibration (seconds): warm median, warm p95 and cold first-query median per query at 1/2/4 vCPU (s1/s2/s4). Source: `results/calib_docker.json`.

query	s1 warm	s1 p95	s1 cold	s2 warm	s2 p95	s2 cold	s4 warm	s4 p95	s4 cold
Q01	0.3735	0.3838	0.4526	0.185	0.1864	0.2219	0.0996	0.1171	0.1222
Q03	0.038	0.0392	0.1893	0.025	0.0266	0.09	0.0261	0.0264	0.0942
Q04	0.0959	0.0976	0.1665	0.0619	0.0623	0.1156	0.0637	0.07	0.1158
Q05	0.0277	0.0321	0.7059	0.02	0.0317	0.1359	0.0185	0.025	0.1099
Q06	0.0507	0.0512	0.1146	0.0302	0.0323	0.0663	0.0199	0.0221	0.049
Q10	0.0764	0.0767	0.152	0.0552	0.0567	0.1066	0.0347	0.035	0.0844
Q12	0.0933	0.1083	0.164	0.0476	0.0494	0.0914	0.0275	0.0279	0.0643
Q14	0.049	0.0504	0.1421	0.0323	0.0324	0.0745	0.0219	0.0231	0.0747

Table 2 Concurrency calibration (closed loop, 120 s per level) and fit latency $= b_q \max(1, an/c)$: $a = 2.584$, $R^2 = 0.956$, CPU per base second 2.147. Source: `results/calib_contention.json`.

size	n	completed	errors	CPU-s	ratio	pred	resid
s2	1	2158	0	229.72	1.0	1.292	-0.292
s2	2	2277	0	240.282	1.91	2.584	-0.674
s2	4	2001	3	240.542	4.398	5.168	-0.77
s2	8	1511	22	241.01	10.927	10.336	0.591

Table 3 Validation v1: sim v1 vs a real 10-min Docker s2 replay (trace seed 4242); frozen bands 25% (p95) and 20% (CPU-s). Source: `results/validation.json`.

metric	real	sim v1	error	pass
p95 latency (s)	2.3358	0.1859	-92.0%	no
CPU-seconds	654.7649	296.2695	-54.8%	no

Table 4 Validation v2 on a new trace (seed 9191); 61 real queries failed (DiskFull). Source: `results/validation_v2.json`.

sim	p95 real	p95 sim	p95 err	CPU-s real	CPU-s sim	CPU-s err	pass
v1	1.6225	0.1864	-88.5%	634.5695	291.48	-54.1%	no
v2	1.6225	2.1838	+34.6%	634.5695	625.8179	-1.4%	no

Table 5 Synthetic burst sweep, sim v1 (means over 10 seeds; latency not validated). p95 target 0.5 s. Source: `results/sweep.parquet`, `results/sweep.json`.

intensity	gap	policy	p95	p99	cost1k	vcpu_s	cold	meets_p95
high	long	P-peak	0.1027	0.1199	0.0358	4.32e+04	0.0	1.0
high	long	P-mean	61.8166	69.2862	0.009	1.08e+04	0.0	0.0
high	long	P-sched	0.1027	0.1199	0.0148	1.782e+04	0.0	1.0
high	long	S-default	9.792	17.457	0.0208	8760.2327	7.0	0.0
high	long	S-tuned	0.176	1.8969	0.025	1.056e+04	0.0	1.0
high	long	S-floor	0.176	1.8969	0.025	1.056e+04	0.0	1.0
high	long	S-prewarm	0.1856	4.495	0.0213	8985.8638	1.0	1.0
high	short	P-peak	0.1027	0.1199	0.0124	4.32e+04	0.0	1.0
high	short	P-mean	0.1853	0.2187	0.0062	2.16e+04	0.0	1.0
high	short	P-sched	0.1027	0.1199	0.0089	3.114e+04	0.0	1.0
high	short	S-default	0.1655	1.4152	0.0214	2.608e+04	0.0	1.0
high	short	S-tuned	0.1655	1.4152	0.0214	2.608e+04	0.0	1.0
high	short	S-floor	0.1655	1.4152	0.0214	2.608e+04	0.0	1.0
high	short	S-prewarm	0.1655	1.4152	0.0214	2.608e+04	0.0	1.0
low	long	P-peak	0.1033	0.1199	0.4281	4.32e+04	0.0	1.0
low	long	P-mean	0.3749	0.4242	0.107	1.08e+04	0.0	1.0
low	long	P-sched	0.1033	0.1199	0.1766	1.782e+04	0.0	1.0
low	long	S-default	3.0002	14.3423	0.1163	4106.3317	7.0	0.0
low	long	S-tuned	0.3737	0.476	0.163	5757.525	0.0	1.0
low	long	S-floor	0.3737	0.476	0.163	5757.525	0.0	1.0
low	long	S-prewarm	0.3764	1.755	0.1198	4229.6753	1.0	1.0
low	short	P-peak	0.1015	0.1199	0.1487	4.32e+04	0.0	1.0
low	short	P-mean	0.3749	0.4235	0.0372	1.08e+04	0.0	1.0
low	short	P-sched	0.1015	0.1199	0.1072	3.114e+04	0.0	1.0
low	short	S-default	0.3736	0.4795	0.1144	1.164e+04	0.0	1.0
low	short	S-tuned	0.3736	0.4795	0.1144	1.164e+04	0.0	1.0
low	short	S-floor	0.3736	0.4795	0.1144	1.164e+04	0.0	1.0
low	short	S-prewarm	0.3736	0.4795	0.1144	1.164e+04	0.0	1.0
mid	long	P-peak	0.1027	0.1199	0.108	4.32e+04	0.0	1.0
mid	long	P-mean	0.5027	0.7767	0.027	1.08e+04	0.0	0.4
mid	long	P-sched	0.1027	0.1199	0.0445	1.782e+04	0.0	1.0
mid	long	S-default	5.0736	15.1834	0.0428	5993.8323	7.0	0.0
mid	long	S-tuned	0.1867	0.4437	0.0553	7746.05	0.0	1.0
mid	long	S-floor	0.1867	0.4437	0.0553	7746.05	0.0	1.0
mid	long	S-prewarm	0.2003	1.7573	0.0441	6183.5006	1.0	1.0
mid	short	P-peak	0.1033	0.1199	0.0372	4.32e+04	0.0	1.0
mid	short	P-mean	0.5072	0.7725	0.0093	1.08e+04	0.0	0.3
mid	short	P-sched	0.1033	0.1199	0.0268	3.114e+04	0.0	1.0
mid	short	S-default	0.1867	0.45	0.0429	1.744e+04	0.0	1.0
mid	short	S-tuned	0.1867	0.45	0.0429	1.744e+04	0.0	1.0
mid	short	S-floor	0.1867	0.45	0.0429	1.744e+04	0.0	1.0
mid	short	S-prewarm	0.1867	0.45	0.0429	1.744e+04	0.0	1.0

Table 6 Cost per 1,000 queries (USD), sim v1 vs sim v2 (headline: v2), seeds 0–9; /P-peak = ratio to P-peak in the same cell; NV = not validated. Source: `results/sweep.parquet`, `results/sweep_v2.parquet`.

intensity	gap	policy	cost1k v1	cost1k v2	/P-peak v1	/P-peak v2	p95 v2 (NV)
high	long	P-peak	0.0358	0.0358	1.0	1.0	0.1285
high	long	P-mean	0.009	0.009	0.25	0.25	597.5216
high	long	P-sched	0.0148	0.0148	0.413	0.413	0.1285
high	long	S-default	0.0208	0.0359	0.579	1.001	33.7881
high	long	S-tuned	0.025	0.031	0.699	0.864	3.73
high	long	S-floor	0.025	0.031	0.699	0.864	3.73
high	long	S-prewarm	0.0213	0.0289	0.594	0.806	8.4747
high	short	P-peak	0.0124	0.0124	1.0	1.0	0.1286
high	short	P-mean	0.0062	0.0062	0.5	0.5	2.2956
high	short	P-sched	0.0089	0.0089	0.721	0.721	0.1286
high	short	S-default	0.0214	0.0269	1.725	2.171	0.9855
high	short	S-tuned	0.0214	0.0269	1.725	2.171	0.9855
high	short	S-floor	0.0214	0.0269	1.725	2.171	0.9855
high	short	S-prewarm	0.0214	0.0269	1.725	2.171	0.9855
low	long	P-peak	0.4281	0.4281	1.0	1.0	0.1052
low	long	P-mean	0.107	0.107	0.25	0.25	0.9919
low	long	P-sched	0.1766	0.1766	0.413	0.413	0.1052
low	long	S-default	0.1163	0.246	0.272	0.575	10.9328
low	long	S-tuned	0.163	0.1942	0.381	0.454	0.5835
low	long	S-floor	0.163	0.1942	0.381	0.454	0.5835
low	long	S-prewarm	0.1198	0.1689	0.28	0.394	0.895
low	short	P-peak	0.1487	0.1487	1.0	1.0	0.1045
low	short	P-mean	0.0372	0.0372	0.25	0.25	0.9947
low	short	P-sched	0.1072	0.1072	0.721	0.721	0.1045
low	short	S-default	0.1144	0.1464	0.77	0.985	0.5491
low	short	S-tuned	0.1144	0.1464	0.77	0.985	0.5491
low	short	S-floor	0.1144	0.1464	0.77	0.985	0.5491
low	short	S-prewarm	0.1144	0.1464	0.77	0.985	0.5491
mid	long	P-peak	0.108	0.108	1.0	1.0	0.1058
mid	long	P-mean	0.027	0.027	0.25	0.25	29.1859
mid	long	P-sched	0.0445	0.0445	0.413	0.413	0.1058
mid	long	S-default	0.0428	0.1046	0.396	0.969	18.9218
mid	long	S-tuned	0.0553	0.0695	0.512	0.644	0.4569
mid	long	S-floor	0.0553	0.0695	0.512	0.644	0.4569
mid	long	S-prewarm	0.0441	0.0676	0.409	0.626	1.0634
mid	short	P-peak	0.0372	0.0372	1.0	1.0	0.1058
mid	short	P-mean	0.0093	0.0093	0.25	0.25	30.0899
mid	short	P-sched	0.0268	0.0268	0.721	0.721	0.1058
mid	short	S-default	0.0429	0.0572	1.153	1.539	0.3773
mid	short	S-tuned	0.0429	0.0572	1.153	1.539	0.3773
mid	short	S-floor	0.0429	0.0572	1.153	1.539	0.3773
mid	short	S-prewarm	0.0429	0.0572	1.153	1.539	0.3773

Table 7 S-tuned (sim v2): timeout chosen on seeds 0–9, scored on held-out seeds 10–19. Source: `results/sweep_v2.parquet`.

cell	tau (s)	cost1k 0-9	cost1k 10-19	/P-peak 10-19	p95 10-19 (NV)
high/long	1800	0.031	0.0307	0.853	3.6943
high/short	300	0.0269	0.0268	2.161	0.9039
low/long	1800	0.1942	0.1982	0.453	0.6066
low/short	300	0.1464	0.1474	0.986	0.5829
mid/long	1800	0.0695	0.0701	0.647	0.4352
mid/short	300	0.0572	0.0577	1.547	0.3719

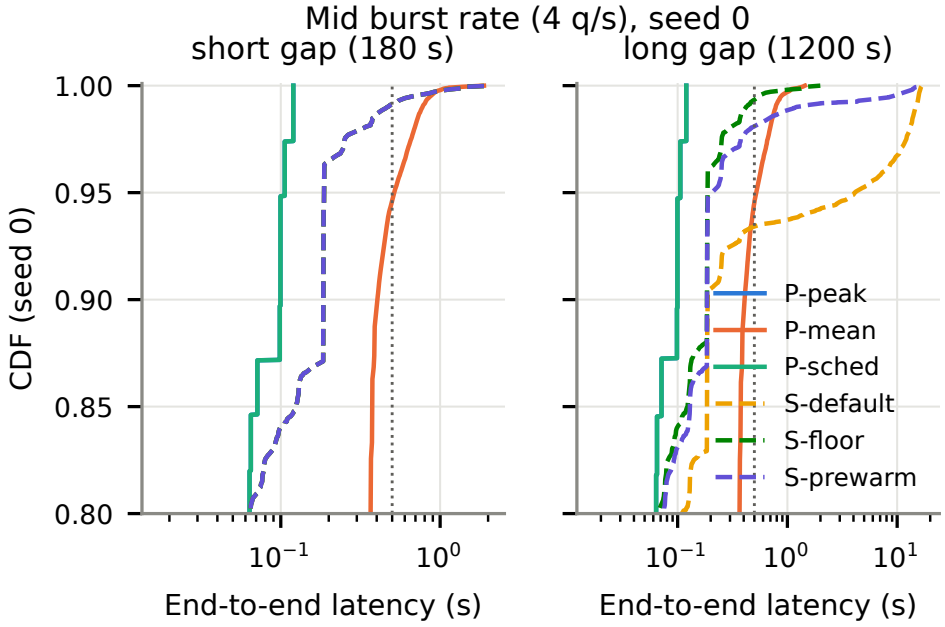


Fig. 5 Placeholder caption. Source: `results/sweep_latency_seed0.parquet`.

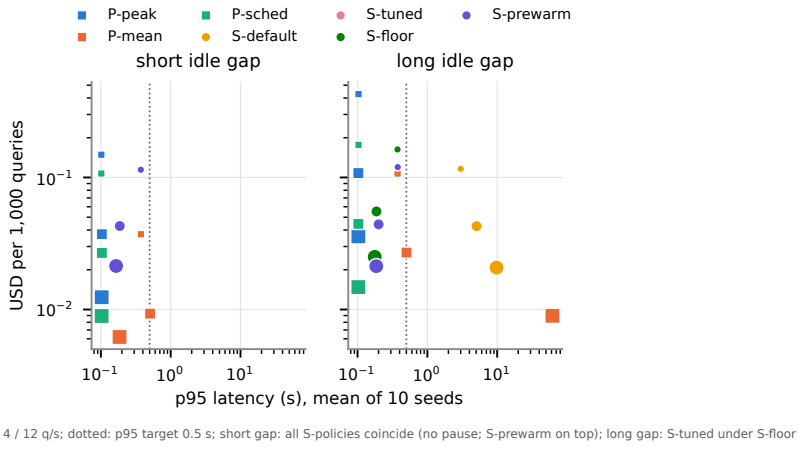


Fig. 6 Placeholder caption. Source: `results/sweep.parquet`.

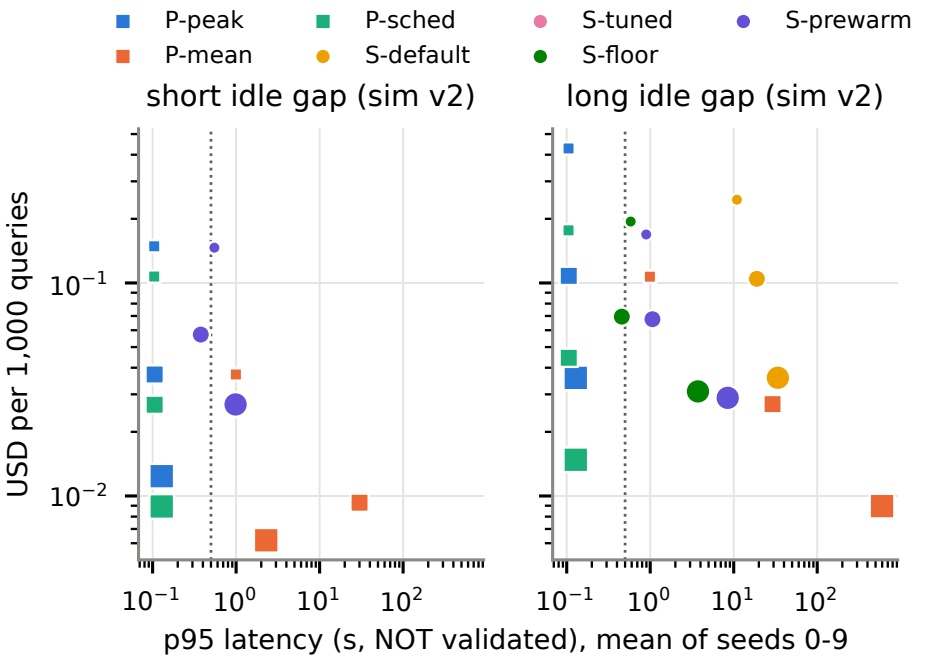


Fig. 7 Placeholder caption. Source: `results/sweep_v2.parquet`.

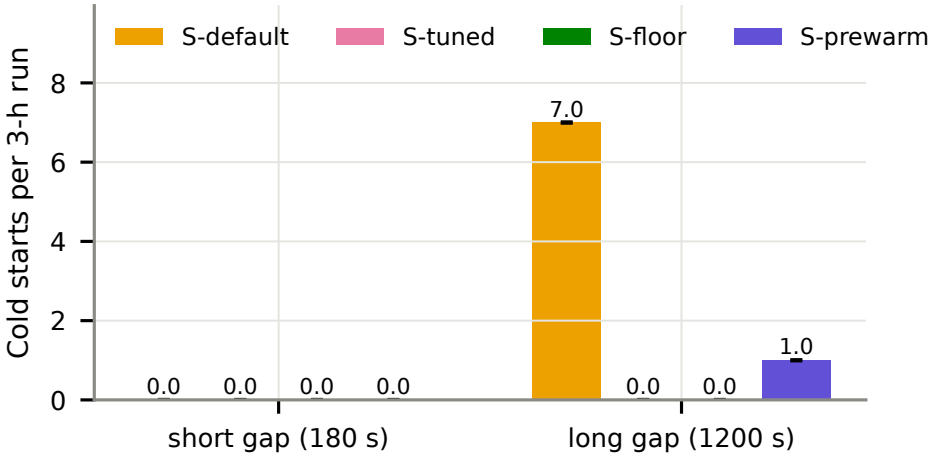


Fig. 8 Placeholder caption. Source: `results/sweep.parquet`.

Table 8 Redset replay, sim v1 (instance 1, 2024-04-10, 28615 queries; means over 10 seeds; latency not validated). Source: `results/redset_sweep.json`, `results/redset_trace.json`.

policy	p95_s	p99_s	compute_vcpu_s	cost_per_1k_usd	cold_starts
P-peak	0.1047	0.1199	3.456e+05	0.2818	0.0
P-mean	1416.9298	1495.5244	8.64e+04	0.0705	0.0
P-sched	0.1047	0.1199	9.72e+04	0.0793	0.0
S-default	0.1152	0.1553	5393.8394	0.0126	1.0
S-tuned	0.1152	0.1553	5152.9394	0.012	1.0
S-floor	0.1116	0.142	2.651e+04	0.0618	0.0
S-prewarm	0.1152	0.1553	5393.8394	0.0126	1.0

Table 9 Redset cost per 1,000 queries, sim v1 vs sim v2; S-tuned τ (v2) = 300 s. Source: `results/redset_sweep.json`, `results/redset_sweep_v2.json`.

policy	cost1k v1	cost1k v2	/P-peak v1	/P-peak v2	p95 v2 (NV)
P-peak	0.2818	0.2818	1.0	1.0	0.2594
P-mean	0.0705	0.0705	0.25	0.25	5754.1977
P-sched	0.0793	0.0793	0.281	0.281	0.2594
S-default	0.0126	0.0139	0.045	0.049	10.7539
S-tuned	0.012	0.0139	0.043	0.049	10.7539
S-floor	0.0618	0.0628	0.219	0.223	0.2656
S-prewarm	0.0126	0.0139	0.045	0.049	10.7539

Table 10 Redset P-mean overload check. Source: `results/redset_overload_check.json`.

quantity	value
P-mean size (vCPU) / servers	1.0 / 1
capacity (q/s)	9.94
rho, day mean	0.033
rho, peak hour (22)	0.799
rho, max minute	3.04
minutes with rho > 1	21
max fluid backlog (queries)	14344.0
backlog drain (s)	1442.3
queue unbounded in overload minutes	True
sim v1 P-mean p95 (s)	1416.9

Table 11 Statistics file index: mean, std, bootstrap 95% CI and Wilcoxon signed-rank vs P-peak (paired by seed) per source, cell, policy and metric. Source: `results/stats.csv`.

source	rows	policies	metrics
redset	42	7	6
redset_v2	42	7	6
redset_v2_heldout	42	7	6
sweep	252	7	6
sweep_v2	252	7	6
sweep_v2_heldout	252	7	6

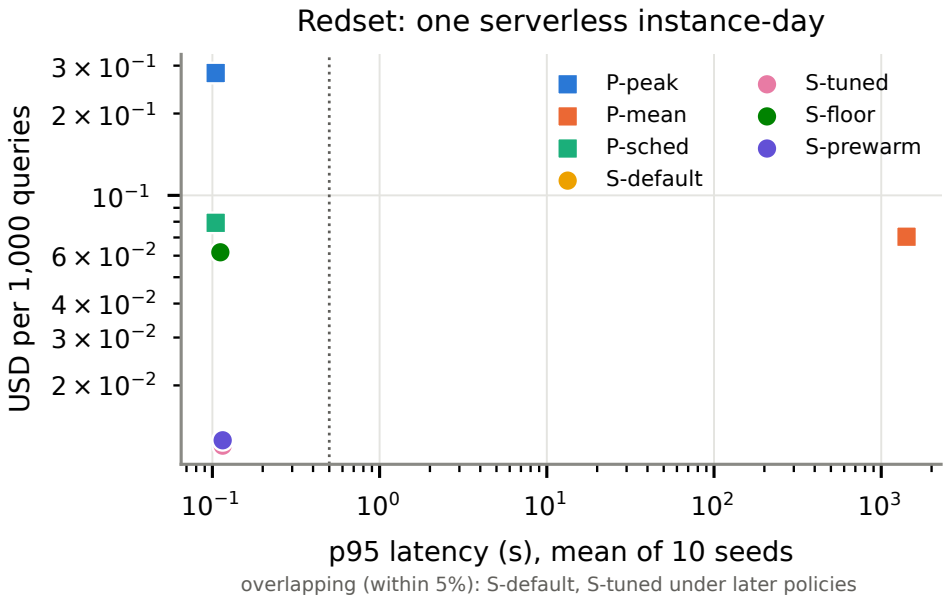


Fig. 9 Placeholder caption. Source: `results/redset_sweep.parquet`.

8 Threats to Validity

9 Conclusion

Declarations

References

[1] Transaction Processing Performance Council: TPC Benchmark H Standard

Specification. Accessed 2026-10-16 (2026). https://www.tpc.org/tpch/

[2] Saxena, G., Rahman, M., Chainani, N., Lin, C., Caragea, G., Chowdhury, F., Marcus, R., Kraska, T., Pandis, I., Narayanaswamy, B.: Why TPC is not enough: An analysis of the Amazon Redshift fleet. Proceedings of the VLDB Endowment **17**(11), 3694–3706 (2024)

[3] Amazon Science: Redset: query metadata from Amazon Redshift (dataset, CC BY-NC 4.0). Accessed 2026-10-16 (2026). https://github.com/amazon-science/redset

[4] Neon: Neon pricing. Accessed 2026-10-16 (2026). https://neon.com/pricing

[5] Amazon Web Services: Amazon Aurora pricing. Accessed 2026-10-16 (2026). https://aws.amazon.com/rds/aurora/pricing/

[6] Amazon Web Services: Scaling to zero ACUs with automatic pause and resume for Aurora serverless. Accessed 2026-10-16 (2026). https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-serverless-v2-auto-pause.html

[7] Amazon Web Services: Amazon RDS for PostgreSQL pricing. Accessed 2026-10-16 (2026). https://aws.amazon.com/rds/postgresql/pricing/

[8] Vantage: db.m7g.large pricing and specs (mirror of the AWS price list). Accessed 2026-10-16 (2026). https://instances.vantage.sh/aws/rds/db.m7g.large

[9] Team SimPy: SimPy: discrete event simulation for Python, version 4.1.2. Accessed 2026-10-16 (2026). https://simpy.readthedocs.io/

[10] The PostgreSQL Global Development Group: PostgreSQL 16 documentation. Accessed 2026-10-16 (2026). https://www.postgresql.org/docs/16/